Available online at www.sciencedirect.com

ScienceDirect

Procedia Computer Science 266 (2025) 1424–1432

Procedia Computer Science

www.elsevier.com/locate/procedia

12th International Conference on Information Technology and Quantitative Management (ITQM 2025)

Robot ground classification based on quaternion transformation and temporal networks

Hanrui Liang*

University of Jinan, Jinan, China

Abstract

In order to enable robots to effectively perceive their environments and maintain high efficiency across various settings, it is essential to propose an innovative ground classification method tailored for robotic systems. Traditional machine learning algorithms rely heavily on feature engineering and often struggle with handling complex datasets. Deep learning algorithms, on the other hand, frequently overlook the impact of spatial pose variations in objects. In this paper, we innovatively proposes a robot ground classification method based on Euler angle transformation and a temporal attention network. Specifically, we first convert the quaternion features from the dataset into Euler angle features. Then, these Euler angles are processed through a long short-term memory (LSTM) neural network and a multi-head attention mechanism to comprehensively extract temporal dependencies and spatial motion features. Finally, the processed features are input into the fully connected layers, achieving a classification accuracy of 87.66%. Comparative experiments demonstrate our method's superiority, and ablation experiments validate the effectiveness of each module.

© 2025 The Authors. Published by Elsevier B.V.
This is an open access article under the CC BY-NC-ND license (https://creativecommons.org/licenses/by-nc-nd/4.0)
Peer-review under responsibility of the scientific committee of the 12th International Conference on Information Technology and Quantitative Management

Keywords: Deep learning; Neural network; Attention mechanism; Long short-term memory

1. Introduction

With the development of science and technology, robot-related technologies and applications have been widely adopted in various scenarios [1], including resource surveying [2], disaster relief [3], military [4] and aerospace [5]. To enable robots to perform tasks with greater stability and efficiency, it is essential that they provide timely feedback based on accurate information about the ground medium. Therefore, developing an algorithm for the automatic detection of a robot's ground medium type holds significant research value.

In the field of artificial intelligence, long short-term memory (LSTM) is capable of processing and predicting time-series data or tasks involving sequential dependencies. The attention mechanism can enhance the weight of key time

* Corresponding author. Tel.: +0-000-000-0000 ; fax: +0-000-000-0000 .
E-mail address: 202331232047@stu.ujn.edu.cn

1877-0509 © 2025 The Authors. Published by Elsevier B.V.
This is an open access article under the CC BY-NC-ND license (https://creativecommons.org/licenses/by-nc-nd/4.0)
Peer-review under responsibility of the scientific committee of the 12th International Conference on Information Technology and Quantitative Management
10.1016/j.procs.2025.08.174

steps, improve training efficiency and classification performance. Given the strong temporal characteristics of signals collected from robots, we combine LSTM with attention mechanisms to address the problem of automatic ground medium type detection.

Specifically, we propose a novel ground detection and classification method based on quaternion angle conversion, LSTM, and attention mechanisms. First, features are extracted from the signal dataset using quaternion angle conversion. Subsequently, LSTM and attention are applied to further extract temporal features. Finally, a fully connected layer is employed to perform ground medium classification. The effectiveness of the proposed method is validated through comparative and ablation experiments, demonstrating its superior performance in ground classification tasks.

2. Literature review

In recent years, ground medium classification methods based on traditional machine learning and deep learning have gained considerable popularity. In traditional machine learning approaches, Yaren Aydin et al. [6] evaluated various algorithms using features such as sieve No. 200 (particle analysis), liquid limit, plastic limit, and plasticity index. Their results showed that tree-based models and gradient boosting ensemble methods perform well even on small datasets. P.C. Dixon et al. [7] utilized three-dimensional accelerometers to extract motion-related features—such as mean, standard deviation, and sample entropy—and trained an extreme gradient boosting (XGBoost) model with promising results. Zuo Jiankai et al. [8] significantly improved the F1 score by proposing a novel hybrid evolutionary learning method. This approach combines naive bayes classifier, support vector machine (SVM), k-nearest neighbors (KNN), decision tree, random forest (RF), and XGBoost in a weighted ensemble, further optimized using an improved genetic algorithm. M. G. Harinarayanan Nampoothiri et al. [9] developed an integrated subspace KNN approach for classifying 11 terrain types, achieving high accuracy. Wenzhao Liao et al. [10] introduced a RF model optimized by a genetic algorithm for ground type classification in complex terrain environments, reaching an accuracy of 93%, significantly outperforming conventional RF models. In the field of deep learning, Wanli Wang et al. [11] proposed a hybrid terrain classification method for mobile robots, which utilizes labeled RGB images to train a model combining convolutional neural networks (CNNs) and SVMs. This approach effectively balances high classification accuracy with limited computational resources. Fabio Vulpi et al. [12] collected proprioceptive data from robots traversing various terrains, applied denoising and normalization preprocessing, and compared the performance of recurrent neural networks (RNNs), CNNs, and SVMs. Their results demonstrated that deep learning models significantly outperformed traditional SVMs. Rudolf J. et al. [13] introduced a continuous classification framework based on LSTM networks, which leverages proprioceptive signals from a hexapod robot to achieve real-time terrain perception with reduced latency compared to conventional SVM approaches. Francesco Lomio et al. [14] used inertial measurement unit (IMU) data as input for ground type classification and achieved an accuracy of 68.21% by integrating XGBoost, fully convolutional networks and ResNet models. Ghaith Al-Refai et al. [15] proposed a cascade fusion framework for enhanced feature extraction, which led to improved detection accuracy.Jixiang Niu et al. [16] applied discrete wavelet transform (DWT) to denoise a Kaggle competition dataset, followed by the Hilbert-Huang Transform (HHT) to extract instantaneous frequencies from the two most significant channels. These features were integrated into the original dataset to expand the feature space, and the data was subsequently processed using CNNs, LSTMs, and attention mechanisms, achieving high classification accuracy. Ziyu Guo [17] employed an enhanced HHT in conjunction with the ensemble empirical mode decomposition (EEMD) algorithm and LSTM networks for ground material classification. The findings revealed that EEMD provided superior accuracy compared to traditional empirical mode decomposition (EMD). Jinhwan Yoo et al. [18] introduced an innovative road surface classification method based on tire–road interaction acoustic signals and CNNs. The method captures acoustic signals generated by tire–road contact, converts them into two-dimensional representations using continuous wavelet transform, and classifies them via CNN and fully connected layers, effectively identifying various road surfaces and tire types. Jianbo Guo et al. [19] proposed a supervised learning approach for multi-source vibration data fusion. This method employs the Mallat algorithm for signal denoising and uses the Gramian Angular Summation Field technique to transform signals into images, which are then overlaid across RGB channels. These composite images are classified using a residual CNN integrated with an attention mechanism, achieving a final classification accuracy of 92.35% for agricultural terrain types.

The aforementioned techniques have improved classification performance. However, they each possess notable limitations. Although traditional machine learning algorithms incur lower training costs, they heavily depend on feature engineering, struggle with complex data, and perform poorly on datasets containing continuous spatial motion features. Deep learning methods based on neural networks can automatically extract data features, but they often overlook the issue of spatial pose variations in objects.

3. Method

As a whole, our model comprises a feature extraction module and a classification module. In the feature extraction module, we first transform the spatial quaternions into euler angles to better represent the robot's motion state features. These are then processed using an LSTM network to capture temporal features, followed by a multi-head attention mechanism to further enhance feature extraction. In the classification module, the final categorization of ground surface types is achieved through fully connected layers, as illustrated in Figure 1.

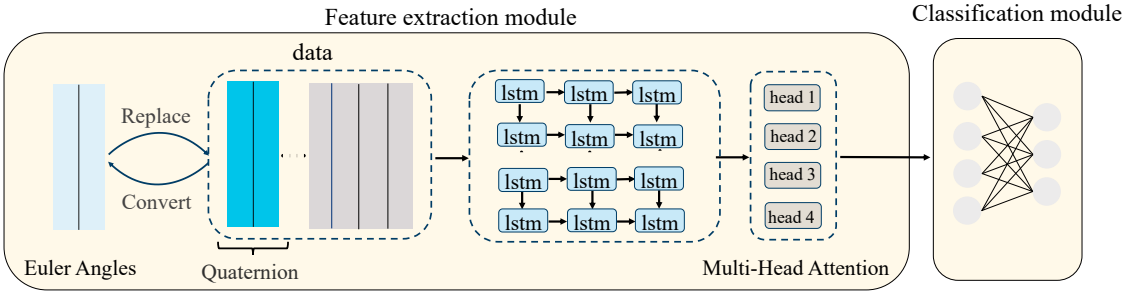


Fig. 1. Surface classification process

3.1. Quaternion transformation

Quaternions are highly sensitive to noise, and even slight perturbations can lead to normalization failure. In contrast, euler angles offer greater tolerance to such disturbances, enabling the model to capture directional features more effectively and train more efficiently. Therefore, we adopt this method for conversion.

Specifically, since the conversion to euler angles implicitly assumes that the input quaternion is a unit quaternion, we first normalize the quaternion, as expressed by the following equation (1) .

$$q = [w, x, y, z] \tag{1}$$

According to equation (2), the norm of the quaternion can be calculated.

$$\| q \| = \sqrt{w^2 + x^2 + y^2 + z^2} \tag{2}$$

The normalization of a quaternion is given in equation (3).

$$q_{\text{normalized}} = \frac{q}{\|q\|} \tag{3}$$

The normalized quaternion is converted to euler angles, which consist of the yaw angle (ψ), pitch angle (θ), and roll angle (ϕ). The conversion formula is shown in equation (4).

$$\begin{bmatrix} \psi \\ \theta \\ \phi \end{bmatrix} = \begin{bmatrix} \arctan\ 2(2(xy + wz), 1 - 2y^2 - 2z^2) \\ \arcsin\ (2(wy - xz)) \\ \arctan\ 2(2(yz + wx), 1 - 2x^2 - 2y^2) \end{bmatrix} \tag{4}$$

3.2. Neural network

3.2.1. LSTM

LSTM is a specialized type of RNN that introduces three gate structures—input gate, forget gate, and output gate—on top of the traditional RNN architecture. By leveraging its gating mechanism and cell state, LSTM can selectively retain or discard information from time series data. Compared to standard RNN, LSTM is more effective at capturing long-term dependencies and mitigating the vanishing gradient problem. In this experiment, each signal group comprises a time sequence of length 128, making LSTM particularly well-suited for the task.

Figure 2 illustrates the structure of LSTM. Let the feature at the current time step be x_t, the output from the previous time step be h_{t-1}, and the cell state be c_t. The input gate, forget gate, and output gate are computed by equations (5), (6), and (7), respectively.

$$i_t = \sigma(W_i \cdot [h_{t-1}, x_t] + b_i) \tag{5}$$

$$f_t = \sigma(W_f \cdot [h_{t-1}, x_t] + b_f) \tag{6}$$

$$o_t = \sigma(W_o \cdot [h_{t-1}, x_t] + b_o) \tag{7}$$

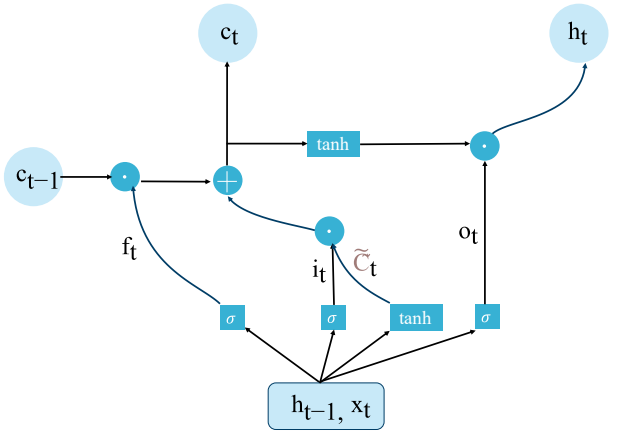


Fig. 2. LSTM unit

The LSTM architecture consists of three stages:

Forgetting phase: The current input x_t and the previous time step output h_{t-1} are fed into the forget gate to compute f_t, which subsequently determines the information to be discarded through the calculation $f_t \odot c_{t-1}$.

Selective memory phase: The extent to which new information is written into the cell state is regulated by the input gate, calculated as $i_t \odot \tilde{C}_t$, where the candidate value $\tilde{C}_t$ is defined by equation (8).

$$\tilde{C}_t = \tanh(W_c \cdot [h_{t-1}, x_t] + b_c) \tag{8}$$

Then, the new cell state is calculated using equation (9).

$$c_t = f_t \odot c_{t-1} + i_t \odot \tilde{C}_t \tag{9}$$

The symbol $\odot$ denotes the element-wise (hadamard) product.

Output phase: The hidden state h_t is obtained by element-wise multiplying the output gate o_t with the cell state c_t, which has been passed through the hyperbolic tangent activation function tanh, as shown in equation (10).

$$h_t = o_t \odot \tanh(c_t) \tag{10}$$

3.2.2. Multi-head attention mechanism

The multi-head attention mechanism is a refinement of the self-attention mechanism, which maps the input time series into multiple low-dimensional subspaces to perform the attention computations in parallel and subsequently concatenates the outputs, which is able to capture diversified feature dependencies in different subspaces and reinforce the weights of key time steps within the sequence. The structure of the multi-head attention mechanism is illustrated in Figure 3.

The input time series is linearly projected to generate the Query(Q), Key(K), and Value(V) matrices. Q denotes the time steps to be attended to, K is used to match the similarity with Q to filter the time step that needs to be concerned, and V contains contains the specific information associated with each relevant time step, which is aggregated through a weighted sum after the attention weights are computed. The formula for multi-head splitting is presented in equation (11).

$$\text{head}_i = \text{Attention}(QW_i^Q, KW_i^K, VW_i^V) \tag{11}$$

W_i^Q, W_i^K, W_i^V are query projection, key projection and value projection.

The scaled dot product attention is computed independently for each head, as shown in equation (12).

$$\text{Attention}(Q, K, V) = \text{softmax}(\frac{QK^\top}{\sqrt{d_k}})V \tag{12}$$

where d_k is the dimension of the vector.

As described in equation (13), the multi-head attention outputs are concatenated and the original dimensions are subsequently restored.

$$\text{MultiHead}(Q, K, V) = \text{Concat}(\text{head}_1, \dots, \text{head}_n)W^O \tag{13}$$

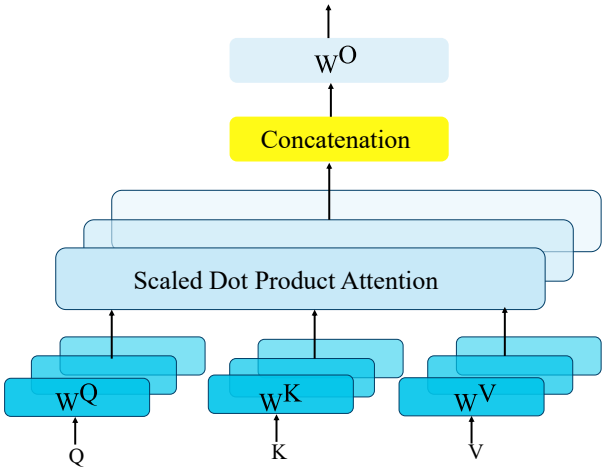


Fig. 3. Multi-Head Attention

4. Experiments

4.1. Introduction to the experimental environment and dataset

This study establishes the experimental framework using the Python 3.9.21 interpreter and the PyTorch 1.12 deep learning framework. Model training was conducted on an NVIDIA GeForce MX250 GPU. The experimental dataset is from the public dataset of CareerCon 2019-Help Navigate Robots competition. The dataset comprises 3,810 time series samples, each containing 128 time steps and 10 features: orientation components (x, y, z, w), angular velocity, and linear acceleration along the x-, y-, and z-axes. The output classification types and corresponding label distributions are illustrated in Figure 4.

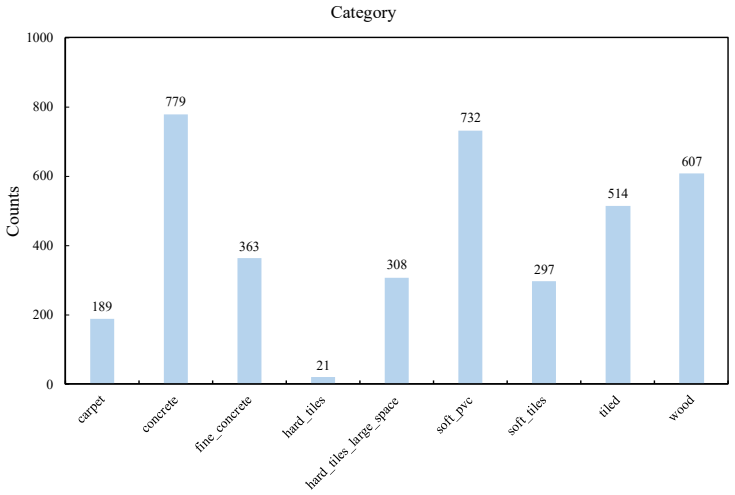


Fig. 4. Number of labels

4.2. Evaluation metrics

To evaluate the model performance in this experiment, we used accuracy, precision, recall, and F1 score for quantification. Accuracy, which measures the proportion of correctly predicted positive and negative instances relative to the total number of predictions, is given by equation (14).

$$\text{Accuracy} = \frac{TP+TN}{TP+FP+FN+TN} \tag{14}$$

Precision, defined as the proportion of correctly predicted positive instances among all instances predicted as positive, is given by equation (15).

$$\text{Precision} = \frac{1}{N}\sum_{i=1}^{N}\frac{TP}{TP+FP} \tag{15}$$

Recall measures the proportion of correctly predicted positive instances to the total number of actual positive instances. It is defined by equation (16).

$$\text{Recall} = \frac{1}{N}\sum_{i=1}^{N}\frac{TP}{TP+FN} \tag{16}$$

The F1 score, which is the harmonic mean of precision and recall, provides a balanced measure between the two. It is defined by equation (17).

$$\text{F1} = 2 \times \frac{Precision \times Recall}{Precision+Recall} \tag{17}$$

Where TP denotes a true positive case, i.e., a positive instance that is correctly predicted as positive; FP denotes a false positive case, i.e., a negative instance incorrectly predicted as positive; FN denotes a false negative case, i.e., a positive instance incorrectly predicted as negative; and TN denotes a true negative case, i.e., a negative instance correctly predicted as negative.

4.3. Hyperparameter settings

In this study, the dataset is split into training and test sets with a ratio of 4:1. The data in the training set is normalized to ensure all features are on the same scale, and the test set is normalized using the same strategy as the training set. During the training process, the AdamW optimizer is used, with an initial learning rate of 0.0003 and a weight decay of 0.0001 to mitigate overfitting. The learning rate scheduling strategy employs cosine annealing with

warm restarts, starting with an initial cycle length of 30 and a cycle length multiplier of 1. This strategy facilitates a better escape from local optima. The remaining optimal hyperparameter settings are summarized in Table 1.

Table 1. The hyperparameters of our model.

Hyperparameters	Value
Batch size	128
Epoch size	200
Number of LSTM units	128
Number of LSTM layers	3
Dropout rate	0.2
Multi-head attention numbers of heads	4
Multi-head attention embedding dimension	128

4.4. Comparative experiments

To evaluate the effectiveness of the proposed method, we conducted a comparative experiment, where our method was compared with several popular deep learning and machine learning models. The results are presented in Table 2. As shown, our method outperforms the RNN model by achieving a relative improvement of 4.04% in precision, and a 6.06% improvement in recall compared to the GPU model. The advantages of our method over traditional machine learning models are even more pronounced: accuracy improved by 6.95% to 13.91%, precision increased by 14.68% to 17.56%, and recall and F1 score improved by 18%. These results demonstrate that the method proposed in this paper significantly outperforms existing models.

Table 2. Comparsion of our method with other methods.

Method	Accuracy(%)	Precision(%)	Recall(%)	F1 score(%)
Our method	87.66	89.92	88.55	89.11
RNN	84.51	85.88	85.81	85.73
GRU	82.55	83.96	81.59	82.54
SVM	73.75	75.24	69.79	71.02
XGBoost	80.71	72.36	71.98	72.12

4.5. Ablation experiments

To validate the effectiveness of the ground detection classification method based on quaternion angle conversion, LSTM, and attention, we conducted an experiment using a combination of modules (with the same hyperparameters as those used in the proposed method). This experiment was designed to demonstrate the contribution of each module. The results are shown in Figure 5.

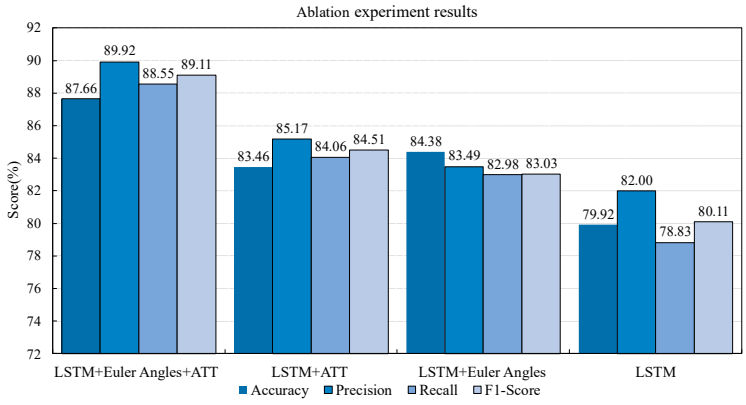


Fig. 5. Ablation experiment results

LSTM classification alone is capable of capturing only the temporal features in the data, which results in a relatively low classification evaluation metrics (Recall = 0.7883). When combined with the attention mechanism, LSTM enhances the weight of certain time steps, leading to an improvement in the evaluation metrics by 3% to 5% compared to LSTM alone. However, this approach still struggles to effectively capture spatial motion state features. The integration of euler angle transformation in our method allows the model to better capture these spatial motion features, thus achieving the highest evaluation metrics.

5. Discussion

Our proposed method achieves superior classification performance on the experimental dataset. Quantitative metrics from comparative experiments demonstrate that it significantly outperforms several widely used baseline methods. Ablation experiments further reveal that transforming quaternions into euler angles yields markedly better results than using the original quaternion data, likely because euler angles provide more interpretable spatial features that are easier for the model to learn. Additionally, incorporating the attention mechanism after the LSTM module enhances the model's ability to assign greater weight to critical time steps, thereby improving the extraction of key temporal information. Compared with prior methods, our approach features a smaller model size, shorter training time, and improved cost-efficiency, effectively balancing classification performance with computational resource constraints. However, the system still has some limitations and could be improved in the following ways:

Signal denoising: IMU sensors inevitably capture external noise during data collection, which adversely affects feature extraction and final classification performance. To address this, DWT can be applied to decompose the signal. Since signal components generally have larger wavelet coefficients while noise tends to have smaller coefficients, appropriate thresholding can effectively denoise the input data.

Integration of temporal and positional information: The dataset includes accelerometer, gyroscope, and orientation data recorded during continuous robot motion. To better integrate both spatial and temporal features, a hybrid model combining CNN and LSTM can be employed. This allows the CNN to capture local spatial dependencies while the LSTM models long-term temporal dynamics.

Acceleration of temporal prediction: To improve prediction speed on temporal data, the original time series can be segmented using a sliding window approach. The hilbert transform can then be used to extract frequency-domain representations from each segment, accelerating temporal modeling and enhancing real-time applicability.

6. Conclusion

In this paper, we propose ground media classification method based on euler angle transformation and a combined LSTM-attention architecture, using the robot ground material classification dataset from Kaggle website as an experimental dataset. We first convert the quaternion features in the dataset into euler angles. These transformed features are then processed using a three-layer LSTM network with 128 units per layer, followed by a four-head attention mechanism. Finally, a fully connected layer is used to perform the classification. In our experiments, we demonstrate the superior classification accuracy (87.66%) of the proposed method through comparative experiments. Moreover, ablation studies validate the effectiveness and advancement of each part of the proposed method, which proves the superiority of euler angle transformation and the joint LSTM-attention mechanism. We contend that how to improve the data preprocessing to enhance the neural network's ability to learn spatial motion features more efficiently and address the problem of data imbalance is a key direction for future research.

References

[1] Raj, R., & Kos, A. (2022). A comprehensive study of mobile robot: History, developments, applications, and future research perspectives. Applied Sciences, 12(14), 6951.

[2] Topolsky, D., Topolskaya, I., Plaksina, I., Shaburov, P., Yumagulov, N., Fedorov, D., & Zvereva, E. (2022). Development of a mobile robot for mine exploration. Processes, 10(5), 865.

[3] Trevelyan, J., Hamel, W. R., & Kang, S. C. (2016). Robotics in hazardous applications. Springer handbook of robotics, 1521-1548.

[4] Lin, P., Bekey, G., & Abney, K. (2008). Autonomous military robotics: Risk, ethics, and design.

[5] Zhang, Y., Li, P., Quan, J., Li, L., Zhang, G., & Zhou, D. (2023). Progress, challenges, and prospects of soft robotics for space applications. Advanced Intelligent Systems, 5(3), 2200071.

[6] Aydın, Y., Işıkdağ, Ü., Bekdaş, G., Nigdeli, S. M., & Geem, Z. W. (2023). Use of machine learning techniques in soil classification. Sustainability, 15(3), 2374.

[7] Dixon, P. C., Schütte, K. H., Vanwanseele, B., Jacobs, J. V., Dennerlein, J. T., Schiffman, J. M., ... & Hu, B. (2019). Machine learning algorithms can classify outdoor terrain types during running using accelerometry data. Gait & posture, 74, 176-181.

[8] Zuo, J., & Zhang, Y. (2021). A hybrid evolutionary learning classification for robot ground pattern recognition. Journal of Intelligent & Fuzzy Systems, 40(5), 10129-10143.

[9] Nampoothiri, M. H., Anand, P. G., & Antony, R. (2020). Real time terrain identification of autonomous robots using machine learning. International Journal of Intelligent Robotics and Applications, 4(3), 265-277.

[10] Liao, W. (2023, April). Ground classification based on optimal random forest model. In 2023 IEEE International Conference on Control, Electronics and Computer Technology (ICCECT) (pp. 709-714). IEEE.

[11] Wang, W., Zhang, B., Wu, K., Chepinskiy, S. A., Zhilenkov, A. A., Chernyi, S., & Krasnov, A. Y. (2022). A visual terrain classification method for mobile robots' navigation based on convolutional neural network and support vector machine. Transactions of the Institute of Measurement and Control, 44(4), 744-753.

[12] Vulpi, F., Milella, A., Marani, R., & Reina, G. (2021). Recurrent and convolutional neural networks for deep terrain classification by autonomous robots. Journal of Terramechanics, 96, 119-131.

[13] Szadkowski, R. J., Drchal, J., & Faigl, J. (2018, September). Terrain classification with crawling robot using long short-term memory network. In International Conference on Artificial Neural Networks (pp. 771-780). Cham: Springer International Publishing.

[14] Lomio, F., Skenderi, E., Mohamadi, D., Collin, J., Ghabcheloo, R., & Huttunen, H. (2019). Surface type classification for autonomous robot indoor navigation. arXiv preprint arXiv:1905.00252.

[15] Al-refai, G., Karasneh, D., Elmoaqet, H., Ryalat, M., & Almtireen, N. (2025). Surface Classification from Robot Internal Measurement Unit Time-Series Data Using Cascaded and Parallel Deep Learning Fusion Models. Machines, 13(3), 251.

[16] Niu, J., Li, H., Liu, Z., Liu, W., & Xu, H. (2023). Robot Ground Media Classification Based on Hilbert–Huang Transform and Attention-Based Spatiotemporal Coupled Network. IET Signal Processing, 2023(1), 4721508.

[17] Guo, Z. (2024). Robot terrain classification based on improved Hilbert-Huang transform and long short-term memory network. Applied and Computational Engineering, 95, 49-56.

[18] Yoo, J., Lee, C. H., Jea, H. M., Lee, S. K., Yoon, Y., Lee, J., ... & Hwang, S. U. (2022). Classification of road surfaces based on CNN architecture and tire acoustical signals. Applied Sciences, 12(19), 9521.

[19] Guo, J., Wang, S., Mao, Y., Wang, G., Wu, G., Wu, Y., & Liu, Z. (2024). Supervised learning study on ground classification and state recognition of agricultural robots based on multi-source vibration data fusion. Computers and Electronics in Agriculture, 219, 108791.